

A MOD 31^2 COUNTEREXAMPLE TO A GENERALIZED CUBIC PARTITION CONGRUENCE OF DAS, MAITY AND SAIKIA

ABSTRACT. For $c \geq 1$ let $a_c(n)$ be the number of partitions of n in which the even parts may take c colours, so that $\sum_{n \geq 0} a_c(n)q^n = 1/(f_1 f_2^{c-1})$ with $f_k = (q^k; q^k)_\infty$. In the Concluding Remarks of the preprint [2], Das, Maity and Saikia conjecture three congruences to a prime-square modulus, the first of which is $a_{25}(31^2 n + 644) \equiv 0 \pmod{31^2}$ for all $n \geq 0$; they state explicitly that they leave it open because the Ramanujan–Kolberg bound is too large for their computer. We refute it at $n = 2$. The exact value of $a_{25}(2566)$ is a 182-digit integer, printed in full below; it satisfies $a_{25}(2566) \equiv 682 = 22 \cdot 31 \pmod{31^2}$, so its 31-adic valuation is exactly 1 and the congruence fails. What survives is the first power: $a_{25}(31^2 n + 644) \equiv 0 \pmod{31}$ does hold for all n , but that is a published theorem of Guadalupe [4], and it is only its lift to 31^2 that is false.

1. THE CONJECTURE

Write $f_k = (q^k; q^k)_\infty$. Following Amdeberhan, Sellers and Singh [1], a *generalized cubic partition* of n is a partition of n in which each *even* part may appear in one of $c \geq 1$ colours, and $a_c(n)$ denotes the number of these, with $a_c(0) = 1$. Equivalently, and this is the definition used throughout,

$$(1) \quad \sum_{n \geq 0} a_c(n)q^n = \frac{1}{f_1 f_2^{c-1}}.$$

Thus $a_1(n)$ is the ordinary partition function (OEIS A000041) and $a_2(n)$ counts Chan’s cubic partitions (OEIS A002513) [5].

Remark 1. The source [2] defines $a_c(n)$ by (1), but its prose reads “each part may appear in $c \geq 1$ different colors”, which would give $1/f_1^c$ instead. The word *even* was dropped; both [1], where the function originates, and [4] carry it. Equation (1) is the authoritative object, and nothing below is sensitive to the choice: under the prose reading $1/f_1^{25}$ the congruence at issue already fails at $n = 0$, with residue 550 rather than 0 modulo 961.

The statement refuted here is the first displayed congruence of the first conjecture in the Concluding Remarks of `arXiv:2503.19399v2` [2], quoted below from the source of that version. We identify it by that position and

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not by a number: no claim is made here about how the source numbers this conjecture, and [3] refers to its third congruence as “Conjecture 7.3”.

Conjecture (Concluding Remarks of [arXiv:2503.19399v2](#) [2]). *For all $n \geq 0$, we have*

$$(2) \quad a_{25}(31^2n + 644) \equiv 0 \pmod{31^2},$$

$$(3) \quad a_{41}(47^2n + 256) \equiv 0 \pmod{47^2},$$

$$(4) \quad a_{61}(67^2n + 555) \equiv 0 \pmod{67^2}.$$

The single quantifier “for all $n \geq 0$ ” scopes the whole display: there is no side hypothesis, no exceptional set and no “sufficiently large n ”, and the constants $c = 25$, $p = 31$ and $r = 644$ are fixed, so one explicit n refutes (2). The authors print Ramanujan–Kolberg parameters for each of the three lines, including the bound $\lfloor \nu \rfloor = 3813$ of Radu’s algorithm [6] for (2), and then write, verbatim:

“Since the bound ν is large for the computation to take a significant amount of time in a modest computer, we leave the above as an open problem.”

Deciding (2) at a single n , however, needs one coefficient of one truncated series and not the cells of a certificate.

2. THE COUNTEREXAMPLE

Theorem 2. *Congruence (2), the first displayed congruence of the conjecture quoted above from [arXiv:2503.19399v2](#) [2], is false. Explicitly it fails at $n = 2$: with $N = 31^2 \cdot 2 + 644 = 1922 + 644 = 2566$,*

$$a_{25}(2566) \equiv 682 = 22 \cdot 31 \pmod{31^2}, \quad \text{so} \quad v_{31}(a_{25}(2566)) = 1.$$

In particular $31 \mid a_{25}(2566)$ while $31^2 \nmid a_{25}(2566)$.

Proof. Let W be the 182-digit integer

1471465862381281557802668516631902844710
2593034654589643689045403133888112437860
2701871704379341548682799083362285352501
0333330703396635284385120674666433997413
0010853309981128071099

and let Q be the 180-digit integer

4746664072197682444524737150425493047452
4493660176095624803372268173832620767291
1941521627030134028009029301168662427422
6881711946440758981887486047311077411009
68092042257455744229

Then $a_{25}(2566) = W$, and $31Q = W$, so $31 \mid W$. Reducing Q modulo 31 gives $22 \neq 0$, whence $v_{31}(W) = 1$ exactly and $W \equiv 31 \cdot 22 = 682 \pmod{961}$.

Since $682 \neq 0$ and $N = 2566 = 31^2 \cdot 2 + 644$ lies on the progression of (2) with $n = 2 \geq 0$, the congruence asserted there fails at that cell.

Only the identity $a_{25}(2566) = W$ needs a computation, and it is verified in §4; the rest is the division of one printed integer by another. A reader with any computer algebra system needs the coefficient of q^{2566} in $1/((q; q)_\infty(q^2; q^2)_\infty^{24})$, a 2567-term truncation, and one division by 31. $\square$

3. THE COMPANION CONGRUENCE MODULO 31

That the whole progression is divisible by the first power of 31 is not new and is not claimed here. Theorem 3 of Guadalupe [4] reads: *let $p \geq 7$ be a prime with $p \equiv 3, 7 \pmod{8}$; then for all $n \geq 0$, $a_{p-6}(pn + 13(p^2 - 1)/24) \equiv 0 \pmod{p}$.* At $p = 31$ we have $31 \equiv 7 \pmod{8}$ and $13(31^2 - 1)/24 = 520 = 31 \cdot 16 + 24$, while $961n + 644 = 31(31n + 20) + 24$ with $31n + 20 \geq 20 \geq 16$ for every $n \geq 0$. Hence

$$(5) \quad a_{25}(31^2n + 644) \equiv 0 \pmod{31} \quad \text{for all } n \geq 0$$

is a published theorem, predating [2] and cited in it. What (2) adds to (5) is the second power, and it is only that lift which Theorem 2 refutes.

4. THE COMPUTATION AND `VERIFY.PY`

The single coefficient that decides Theorem 2 is the 2567th of a truncated q -series, so the computation is small; what needs care is that the right series is being expanded. The program `verify.py` accompanying this note is Python 3.9 or later, standard library only, exact integer arithmetic, no external data file. It recomputes $a_{25}(0), \dots, a_{25}(2566)$ over $\mathbb{Z}$ by two routes that share no arithmetic — (i) building $D = f_1 f_2^{24}$ from the generalized pentagonal numbers and inverting it by $A_0 = 1$, $A_n = -\sum_{j=1}^n D_j A_{n-j}$, in which no division and no modular inverse occurs because $D_0 = 1$; and (ii) the logarithmic-derivative recurrence $n a_c(n) = \sum_{j=1}^n B(j) a_c(n-j)$ with $B(j) = \sigma(j) + 2(c-1)\sigma(j/2)[2 \mid j]$, which forms no q -product and touches no pentagonal number, every division by n being asserted exact — finds the two routes equal at all 2567 coefficients, and compares the result against the integers W and Q of §2 *as decimal strings*, so a misprint in this paper would fail the program.

Two published sequences pin (1) against Remark 1: at $c = 1$ the program reproduces OEIS A000041 and at $c = 2$ OEIS A002513 [5]. The recorded run of `verify.py` reports

VERDICT: ALL 35 CHECKS PASS

and the program exits nonzero if any check fails. The run reports more than this note uses: residue-class sweeps, checks that congruences proved in [2] are not violated over the bounded ranges swept, and residues for the other two displayed lines. None of that is claimed here, and all of it is a statement about a bounded range. Its output also records what it does not do: it proves none of the three congruences on any progression, it does

not compute the exact integers behind the other two lines, and it does not check the source’s Ramanujan–Kolberg parameter table against [6] — its final section recomputes one set of residues from the formula as printed in the source, and this note makes no claim about that table.

5. SCOPE

Only (2), as printed and quoted in §1, is refuted, and only by exhibiting one n . Nothing here is a claim about v_{31} at an unlisted cell: the congruence does hold at $n = 0, 1, 10$ of the window checked by `verify.py`, with $v_{31} = 2$ exactly at $n = 0$ and $n = 1$, and no assertion is made about how often that happens.

Nothing new modulo the first power of 31 is proved here: (5) is the $p = 31$ case of Theorem 3 of [4] and is quoted, not proved.

No Ramanujan–Kolberg certificate is offered or completed, in either direction: nothing checked here would have proved (2) had the cells examined all vanished, and Theorem 2 needs no certificate.

Finally, all quotations and parameter values are taken from the source of `arXiv:2503.19399v2`, submitted 28 April 2026, and the wording quoted in §1 is established here for that version only. The version of record (*Eur. J. Math.* **12** (2026), no. 3, Paper No. 34) was revised on 26 May 2026 and accepted on 28 May 2026, that is, after that `arXiv` version; its body is paywalled and was not read, so whether the accepted version prints this conjecture in this wording is not established here.

REFERENCES

- [1] T. Amdeberhan, J. A. Sellers, and A. Singh, *Arithmetic properties for generalized cubic partitions and overpartitions modulo a prime*, *Aequationes Math.* **99** (2025), 1197–1208. doi:10.1007/s00010-024-01116-7. The definition of $a_c(n)$ originates here, with the word *even*.
- [2] H. Das, S. Maity, and M. P. Saikia, *Arithmetic properties of generalized cubic and overcubic partitions*, `arXiv:2503.19399v2` (28 April 2026); published as *Eur. J. Math.* **12** (2026), no. 3, Paper No. 34, 22 pp. The conjecture quoted here, and every quotation and parameter value, are from the `arXiv` source of v2; the journal text was not consulted.
- [3] S. Ghoshal and A. Jana, *Combinatorial proof of a result on generalized overcubic partitions and related conjectures*, `arXiv:2512.04775` (2025). Quotes the third congruence of the same conjecture of [2] as “Conjecture 7.3”.
- [4] R. Guadalupe, *A note on congruences for generalized cubic partitions modulo primes*, *Integers* **25** (2025), Article A20; available from the journal. Theorem 3 gives (5) at $p = 31$.
- [5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequences A000041 (a_1 , the partition function) and A002513 (a_2 , the cubic partitions).
- [6] S. Radu, *An algorithmic approach to Ramanujan’s congruences*, *Ramanujan J.* **20** (2009), 215–251. doi:10.1007/s11139-009-9174-0. Not obtained; cited only for the algorithm whose bound is reported in [2].